A proof of Seymour's second-neighborhood conjecture

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Abstract

We prove that every nonempty finite oriented graph has a vertex with at least as many vertices at directed distance two as at directed distance one. This resolves Seymour's second neighborhood conjecture positively.

1 Introduction

The result. An *oriented graph* is a finite directed graph with no loops, no multiple arcs, and no pair of opposite arcs; it need not be complete or connected. We write $x \to y$ for an arc. For a vertex v put

$$N_1^+(v) = \{u : v \to u\}, \qquad N_2^+(v) = \{w \notin \{v\} \cup N_1^+(v) : v \to u \to w \text{ for some } u\},$$

so $N_2^+(v)$ is the set of vertices at directed distance exactly two from v, each counted once.

Theorem 1.1. *Every nonempty finite oriented graph has a vertex v such that $|N_1^+(v)| \le |N_2^+(v)|$.*

This is Seymour's second neighborhood conjecture, with no connectivity or degree hypothesis. A *sink* (a vertex with $N_1^+(v) = \varnothing$) has $N_2^+(v) = \varnothing$ and satisfies the conclusion. Section 5 derives weighted forms through Seacrest's equivalences [19], and Section 6 derives consequences for short directed cycles.

Image operator. For a digraph D and $S \subseteq V(D)$ define

$$F_D(S) = \{y : x \to y \text{ for some } x \in S\}, \qquad F_D^2(S) = F_D(F_D(S)).$$

F_D does not subtract S; it is monotone and $F_D(A \cup B) = F_D(A) \cup F_D(B)$. In an oriented graph a walk $v \to u \to v$ would use two opposite arcs, so $v \notin F_D^2(\{v\})$ and

$$N_2^+(v) = F_D^2(\{v\}) \setminus F_D(\{v\}). \tag{1}$$

Context. The conjecture was recorded by Dean and Latka [8]. For *tournaments* (an arc between every two vertices) it was proved by Fisher [12] and, combinatorially, by Havet and Thomassé [14]; tournaments missing few arcs are treated in [11, 13, 7, 6]. Small minimum outdegree is handled in [16, 18], approximate ratios (0.657298, then 0.715538) in [5, 15], and random or dense cases in [2, 10, 4]. The proof below builds on the minimal-counterexample and graph-product methods of [3] and on Seacrest's subset perspective [20]. Its new ingredient is a pruning bound for arbitrary finite binary relations (Lemma 3.1).

Proof at a glance. Assume the theorem fails.

1. *Subset deficit (Lemma 2.2).* Take a counterexample D with the fewest vertices and then the fewest arcs. Deleting well-chosen arcs and invoking minimality gives $|F_D^2(S) \setminus F_D(S)| < |F_D(S) \setminus S|$ for every nonempty proper vertex set S.

2. *Blow-up (Lemma 2.3).* Replacing every vertex of D by a large transitive tournament gives an oriented graph G with positive indegrees and

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad (\varnothing \neq U \subsetneq V(G)). \tag{2}$$

   Steps 1–2 form Proposition 2.1.

3. *Pruning (Lemma 3.1).* This step is independent of steps 1–2 and holds for any finite binary relation. Two families $\mathcal{R}, \mathcal{C}$ of ordered pairs, with a "conflict" relation between them, can be made conflict-free by deleting at most $|H| + q$ members. Here H is a set determined by the conflicts and q counts certain points left uncovered by the survivors. The proof bounds a maximum matching in an auxiliary bipartite graph by a rank computation (using Lemma 3.2) and then applies Kőnig's theorem.

4. *Extremal contradiction (Proposition 4.1).* In $X \times X$, $X = V(G)$, consider pairs of families (P, Q) that contain the diagonal and have disjoint coordinate images. Maximize $M + d$, where $M = |P| + |Q|$ and d is the number of points in neither image. Applying (2) to the columns of P and the rows of Q, and taking complements of second images, gives families of total size greater than $M + 2d$. They may conflict. Pruning deletes at most $d + d'$ members, where d' is the new number of uncovered points, so the new objective exceeds $M + 2d - (d + d') + d' = M + d$, a contradiction. This is why the pruning cost must be charged partly to points uncovered *after* deletion.

Dependencies: Lemma 2.2 ⇒ Lemma 2.3 ⇒ Proposition 2.1; Lemma 3.2 ⇒ Lemma 3.1 ⇒ Proposition 4.1; Propositions 2.1 and 4.1 ⇒ Theorem 1.1 ⇒ Corollaries 5.1 (with [19]) and 6.1.

Notation. Letters such as S, Q, T, v have the meaning fixed in the current section. To avoid clashes, five symbols differ from the source: the label sets in Lemma 2.3 are Y_v (source: T_v); the matrix, its support and its index sets in Lemma 3.2 are W, Ω, I_0, J_0 (source: D, T, I, J), with $W^{p,i}, \Omega^{p,i}$ accordingly; the summation indices in (16) are ρ, γ (source: a, b); the matrix in Corollary 5.1 is Φ_D (source: M_D); the non-neighbour set in Corollary 6.1 is $\overline{N}(v)$ (source: $Z(v)$).

2 From a counterexample to a strict subset inequality

In this section $F = F_D$ for the base graph D under discussion.

Proposition 2.1 (Counterexample reduction)**.** *If Theorem 1.1 has a counterexample, then there is a finite nonempty oriented graph G in which every vertex has positive indegree and*

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \textit{for every } \varnothing \neq U \subsetneq V(G). \tag{3}$$

The proof, at the end of this section, combines the two lemmas below. The arc-deletion argument of Lemma 2.2 adapts the proof idea of [20, Lemma 4]. The source reports that, with Seacrest's parameter $\lambda = 1$, that lemma gives the weaker bound $|F^2(S) \setminus F(S)| < |F(S)|$ in an

edge-minimal counterexample when $F(S) \neq \varnothing$. Lemma 2.2 is proved here in full, so nothing depends on that citation. The construction in Lemma 2.3 is the lexicographic product $D[T_m]$ with a transitive tournament T_m, as in [3, Theorem 4.3]; here an estimate is needed for every nonempty proper subset, not only for single vertices.

Lemma 2.2 (Subset deficit)**.** *Call a finite nonempty oriented graph a* counterexample *if* $|N_1^+(v)| > |N_2^+(v)|$ *for every vertex* v*. Suppose counterexamples exist, and let* D *be one of minimum order and, among those, with the minimum number of arcs. Then* D *is strongly connected, every vertex has positive indegree, and*

$$|F^2(S) \setminus F(S)| < |F(S) \setminus S| \qquad \text{for every } \varnothing \neq S \subsetneq V(D). \tag{4}$$

Proof. Plan. Split $F(S)$ into $I = F(S) \cap S$ and $E = F(S) \setminus S$, and measure $g(T) = |F(T) \setminus (I \cup T)|$ for $T \subseteq E$. We grow T from $\varnothing$ to E in pieces Q, showing each time that g increases by less than $|Q|$. Each piece comes from deleting arcs and applying minimality. Subscripts D, D_T on N_1^+, N_2^+ indicate the graph.

Step 1: D is strongly connected with positive indegrees. A sink has $|N_1^+| = 0 = |N_2^+|$, which violates the strict inequality defining a counterexample. So D has no sink, and hence at least two vertices. Suppose D is not strongly connected. Then some strongly connected component $V_0 \subsetneq V(D)$ has no arc leaving it. For $v \in V_0$ every walk from v stays in V_0, so $N_1^+(v)$ and $N_2^+(v)$ are the same in the induced graph $D[V_0]$ as in D. Thus $D[V_0]$ is a counterexample of smaller order, contradicting the choice of D. Being strongly connected with at least two vertices, D has an in-neighbour at every vertex.

Step 2: set-up and the increment claim. Fix $\varnothing \neq S \subsetneq V(D)$ and put

$$I = F(S) \cap S, \qquad E = F(S) \setminus S, \qquad g(T) = |F(T) \setminus (I \cup T)| \quad (T \subseteq E).$$

Thus $F(S) = I \sqcup E$, $I \subseteq S$ and $E \cap S = \varnothing$. Strong connectivity gives an arc from S to its complement, so $E \neq \varnothing$.

Claim. For every $T \subsetneq E$ there is a nonempty $Q \subseteq E \setminus T$ with $g(T \cup Q) - g(T) < |Q|$.

Steps 3–8 prove the Claim for one fixed T; the objects D_T, v, Q, C introduced there depend on T.

Step 3: the deleted graph and the vertex v. Let D_T be D with all arcs from S to $E \setminus T$ deleted. The set $E \setminus T$ is nonempty and each of its vertices lies in $F(S)$, so at least one arc is deleted. Hence D_T is an oriented graph of the same order with fewer arcs, and by the choice of D it is not a counterexample. So some vertex v satisfies

$$|N^+_{1,D_T}(v)| \leq |N^+_{2,D_T}(v)|. \tag{5}$$

Suppose v lost no out-neighbour, i.e. $N^+_{1,D_T}(v) = N^+_{1,D}(v)$. Arcs of D_T are arcs of D, so $F^2_{D_T}(\{v\}) \subseteq F^2_D(\{v\})$, and (1) in both graphs gives $N^+_{2,D_T}(v) \subseteq N^+_{2,D}(v)$. With (5) this yields $|N^+_{1,D}(v)| \leq |N^+_{2,D}(v)|$, contradicting that D is a counterexample. So v lost an out-neighbour. Only arcs from S to $E \setminus T$ were deleted, hence $v \in S$ and the set of lost out-neighbours

$$Q = F(\{v\}) \cap (E \setminus T)$$

is nonempty.

Step 4: where the out-neighbours of v lie. Since $v \in S$, $N^+_{1,D}(v) \subseteq F(S) = I \cup E$, and by definition of Q,

$$N^+_{1,D}(v) \subseteq I \cup T \cup Q, \qquad N^+_{1,D_T}(v) = N^+_{1,D}(v) \setminus Q \subseteq I \cup T, \qquad Q \cap (I \cup T) = \varnothing.$$

Step 5: two-step endpoints in D_T. We show

$$F^2_{D_T}(\{v\}) \subseteq I \cup T \cup F(T). \tag{6}$$

Let $v \to u \to w$ in D_T; by Step 4, $u \in I \cup T$. If $u \in I \subseteq S$, then $w \in F(S) = I \cup E$, and $w \notin E \setminus T$ because arcs from S to $E \setminus T$ were deleted; so $w \in I \cup T$. If $u \in T$, then $w \in F(T)$.

Step 6: comparing second neighbourhoods. Define

$$C = F(Q) \setminus (I \cup T \cup Q \cup F(T)).$$

(i) $C \subseteq N^+_{2,D}(v)$. For $c \in C$ there is a path $v \to q \to c$ in D with $q \in Q$. Step 4 gives $c \notin N^+_{1,D}(v)$, and $c \neq v$ because an arc $q \to v$ would oppose $v \to q$.

(ii) $C \cap N^+_{2,D_T}(v) = \varnothing$, by (6) and the definition of C.

(iii) $N^+_{2,D_T}(v) \setminus N^+_{2,D}(v) \subseteq Q \cap F(T)$. Let w lie in the left side. Then $w \in F^2_{D_T}(\{v\}) \subseteq F^2_D(\{v\})$ and $w \notin N^+_{2,D}(v)$, so (1) gives $w \in N^+_{1,D}(v)$. Also $w \notin N^+_{1,D_T}(v)$, since $w \in N^+_{2,D_T}(v)$. Hence $w \in Q$. By (6) and $Q \cap (I \cup T) = \varnothing$, also $w \in F(T)$.

Together, $N^+_{2,D_T}(v) \subseteq (N^+_{2,D}(v) \setminus C) \cup (Q \cap F(T))$.

Step 7: counting. Write $d_1 = |N^+_{1,D}(v)|$ and $d_2 = |N^+_{2,D}(v)|$, so $d_2 < d_1$. Step 4, (5) and Step 6 (with (i) for the cardinality of $N^+_{2,D}(v) \setminus C$) give

$$d_1 - |Q| = |N^+_{1,D_T}(v)| \leq |N^+_{2,D_T}(v)| \leq d_2 - |C| + |Q \cap F(T)|.$$

Step 8: the change in g is exact. Since $F(T \cup Q) = F(T) \cup F(Q)$, the set $F(T \cup Q) \setminus (I \cup T \cup Q)$ is the disjoint union of $F(T) \setminus (I \cup T \cup Q)$ and $F(Q) \setminus (I \cup T \cup Q \cup F(T)) = C$. The first set is $F(T) \setminus (I \cup T)$ with $Q \cap F(T)$ removed. As $Q \cap (I \cup T) = \varnothing$, we have $Q \cap F(T) \subseteq F(T) \setminus (I \cup T)$, so the first set has $g(T) - |Q \cap F(T)|$ elements. With Step 7,

$$g(T \cup Q) - g(T) = |C| - |Q \cap F(T)| \leq |Q| + d_2 - d_1 < |Q|. \tag{7}$$

This proves the Claim.

Step 9: telescoping. Put $T_0 = \varnothing$. While $T_k \neq E$, apply the Claim to T_k to get Q_k and set $T_{k+1} = T_k \cup Q_k$. Each application deletes arcs from the original D, and the vertex v may change with k. The Q_k are nonempty, pairwise disjoint subsets of E, so the process stops at some $T_r = E$ with $r \geq 1$ and $\sum_k |Q_k| = |E|$. Summing (7) and using $g(\varnothing) = 0$ gives $g(E) < |E| = |F(S) \setminus S|$.

Step 10: identifying $g(E)$. $F(S) = I \cup E$ and $F(I) \subseteq F(S)$ (as $I \subseteq S$), so

$$F^2(S) \setminus F(S) = (F(I) \cup F(E)) \setminus (I \cup E) = F(E) \setminus (I \cup E),$$

whose cardinality is $g(E)$. This proves (4). □

Lemma 2.3 (Transitive-tournament blowup)**.** *Let D be an oriented graph on $n \geq 1$ vertices, all of positive indegree, satisfying* (4) *for every nonempty proper subset of $V(D)$. For an integer $m > n$, replace each vertex v by a transitive tournament on m vertices (the* block *at v). For every arc $v \to w$ of D, put all arcs from the block at v to the block at w, and put no other arcs between distinct blocks. The resulting graph G is oriented, all its indegrees are positive, and it satisfies* (3)*.*

Proof. Plan. Describe $U \subseteq V(G)$ block by block. Each unit of the base deficit (4) contributes $-m$, and the within-block contributions cost at most 1 per base vertex, so $m > n$ wins.

Step 1: inside one block. Label the vertices of each block $1, \dots, m$ with arcs from smaller to larger labels. For a set Y of labels let $f(Y)$ be the set of labels receiving an arc from a label in Y. Then $f(\varnothing) = \varnothing$, and if $Y \neq \varnothing$ has least element a, then $f(Y) = \{a+1, \dots, m\}$ and $f^2(Y) = \{a+2, \dots, m\}$. Since $Y \subseteq \{a, \dots, m\}$,

$$|Y| - |f(Y)| \leq 1, \qquad f^2(Y) \subseteq f(Y), \qquad |Y| - 2|f(Y)| + |f^2(Y)| \leq 1. \tag{8}$$

Step 2: G is oriented with positive indegrees. Within a block the arcs form a tournament. Between two blocks all arcs point one way, because D has no opposite arcs. Each $w \in V(D)$ has an in-neighbour v, and every vertex of the block at w receives arcs from the block at v.

Step 3: notation. Fix $\varnothing \neq U \subsetneq V(G)$. Let Y_v be the set of labels of U in the block at v, and

$$S = \{v \in V(D) : Y_v \neq \varnothing\} \neq \varnothing, \qquad H_0 = F(S) \cup F^2(S).$$

Step 4: the first image, exactly. A vertex in the block at w receives an arc from U in another block exactly when $w \in F(S)$, and then so does the whole block. It receives an arc from U inside its own block exactly when its label lies in $f(Y_w)$. Hence $F_G(U)$ consists of the full blocks indexed by $F(S)$ together with the labels $f(Y_w)$ in each block $w \notin F(S)$ (empty if $w \notin S$), and

$$|F_G(U)| = m|F(S)| + \sum_{v \in S \setminus F(S)} |f(Y_v)|.$$

Step 5: the second image, from above. Apply the description of Step 4 to $U_1 = F_G(U)$. The blocks meeting U_1 are indexed by a set $S_1 \subseteq F(S) \cup S$, so $F(S_1) \subseteq H_0$. In a block $w \notin H_0$ we have $w \notin F(S)$, so U_1 has labels $f(Y_w)$ there and $F_G(U_1)$ has labels $f^2(Y_w)$, empty unless $w \in S$. Bounding the blocks indexed by H_0 by m each,

$$|F_G^2(U)| \leq m|H_0| + \sum_{v \in S \setminus H_0} |f^2(Y_v)|.$$

Step 6: combining, when $S \neq V(D)$. From Steps 4–5 and $|U| = \sum_{v \in S} |Y_v|$,

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| \leq{}& m\big(|H_0| - 2|F(S)|\big) + \sum_{v \in S} |Y_v| \\ & - 2 \sum_{v \in S \setminus F(S)} |f(Y_v)| + \sum_{v \in S \setminus H_0} |f^2(Y_v)|. \end{aligned} \tag{9}$$

Since $F(S) \subseteq H_0$, extending the last sum to $S \setminus F(S)$ adds only nonnegative terms. Now use $|Y_v| \leq m$ for $v \in S \cap F(S)$ and the third inequality of (8) for each $v \in S \setminus F(S)$:

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| &\leq m\big(|H_0| - 2|F(S)| + |S \cap F(S)|\big) + |S \setminus F(S)| \\ &\leq m\big(|F^2(S) \setminus F(S)| - |F(S) \setminus S|\big) + n. \end{aligned}$$

The last line uses $|H_0| = |F(S)| + |F^2(S) \setminus F(S)|$, $|F(S)| - |S \cap F(S)| = |F(S) \setminus S|$ and $|S \setminus F(S)| \le n$. If $S \subsetneq V(D)$, the bracket is a negative integer by (4), so the bound is at most $-m + n < 0$.

Step 7: the case $S = V(D)$. Here (4) is unavailable. Positive indegrees in D give $F(S) = V(D)$, so $F_G(U) = V(G)$ by Step 4. Positive indegrees in G give $F_G^2(U) = F_G(V(G)) = V(G)$. Hence $|U| - 2|F_G(U)| + |F_G^2(U)| = |U| - |V(G)| < 0$, as U is proper. □

Proof of Proposition 2.1. Choose a counterexample D of minimum order and then minimum number of arcs. Lemma 2.2 gives positive indegrees and (4). Lemma 2.3 with any integer $m > |V(D)|$ then gives G. □

3 Pruning two families of pairs

This section is independent of Section 2; no orientation assumption is made. Lemma 3.1 is used once, in the last step of Proposition 4.1.

Lemma 3.1 (Pruning)**.** *Let X be a finite set with a binary relation denoted by $\to$, and let $\mathcal{R}, \mathcal{C} \subseteq X \times X$ be two families, regarded as separate copies even when their coordinates agree. Say that $(p, j) \in \mathcal{R}$ and $(i, s) \in \mathcal{C}$* conflict *if*

$$p \to i \qquad \text{and} \qquad s \to j.$$

Define

$$Z = \{(i, j) : \text{some } (p, j) \in \mathcal{R} \text{ conflicts with some } (i, s) \in \mathcal{C}\},$$
$$H = \{(p, s) : \text{some } (p, j) \in \mathcal{R} \text{ conflicts with some } (i, s) \in \mathcal{C}\}.$$

A point (i, j) is covered *by a member $(p, j) \in \mathcal{R}$ when $p \to i$, and by a member $(i, s) \in \mathcal{C}$ when $s \to j$; these are the only ways a member covers a point.*

There are subfamilies $\mathcal{R}' \subseteq \mathcal{R}$ and $\mathcal{C}' \subseteq \mathcal{C}$ with no conflicts between them such that every initially conflict-free member is retained and

$$|\mathcal{R} \setminus \mathcal{R}'| + |\mathcal{C} \setminus \mathcal{C}'| \le |H| + |\{z \in Z : z \text{ is covered by no member of } \mathcal{R}' \cup \mathcal{C}'\}|. \tag{10}$$

The two sides are counted separately throughout.

Sanity check. If $\mathcal{R} = \{(p, j_1), (p, j_2)\}$, $\mathcal{C} = \{(i_1, s), (i_2, s)\}$ and all four pairs conflict, then $|H| = 1$, $|Z| = 4$, and one whole side must go. Deleting only $\mathcal{R}$ costs $2 > 1 + 0$, but deleting everything costs $4 \le 1 + 4$. The uncovered term is essential.

The next lemma is the linear-algebra fact used in Part G below. A real matrix with rows indexed by I_0 and columns by J_0 is *supported on* $\Omega \subseteq I_0 \times J_0$ if its entries outside Ω are zero; entries in Ω may vanish.

Lemma 3.2 (Kernels at maximum support rank)**.** *Let $\Omega \subseteq I_0 \times J_0$ with I_0, J_0 finite. Suppose that $W : \mathbb{R}^{J_0} \to \mathbb{R}^{I_0}$ has maximum rank among all real matrices supported on Ω. Then for every $x \in \ker W$, every $y \in \ker W^{\mathsf{T}}$, and every $(s, j) \in \Omega$, one has $x_j y_s = 0$.*

Proof. Suppose $x_j y_s \neq 0$, and let $W_1 = W + e_s e_j^{\mathsf{T}}$, which is supported on Ω. Then $W_1 x = Wx + x_j e_s = x_j e_s$, so $e_s \in \operatorname{im} W_1$. For every $w \in \mathbb{R}^{J_0}$, $Ww = W_1 w - w_j e_s \in \operatorname{im} W_1$, so $\operatorname{im} W + \mathbb{R} e_s \subseteq \operatorname{im} W_1$. Since $y^{\mathsf{T}} W = 0$, the vector y is orthogonal to $\operatorname{im} W$, while $y \cdot e_s = y_s \neq 0$; hence $e_s \notin \operatorname{im} W$. Therefore $\operatorname{rank} W_1 \ge \operatorname{rank} W + 1$, contradicting maximality. □

Proof of Lemma 3.1. Plan. Part A turns the lemma into a bound on vertex covers of an auxiliary bipartite graph Γ. Parts B–H show that every matching of Γ has at most $|Z| + |H|$ edges, by a rank computation with four matrices satisfying $NL = AB$. Part I converts a maximum matching into the cover. If there are no conflicts, keep every member ($Z = H = \varnothing$). Vectors are written in coordinate bases $e_\bullet$, and an empty determinant equals 1.

Part A: the auxiliary graph and the reduction. Let Z_L, Z_R be disjoint copies of Z, with z_L, z_R the copies of z. Let Γ be the bipartite graph with sides

$$\mathcal{V}_L = \mathcal{R} \sqcup Z_L, \qquad \mathcal{V}_R = \mathcal{C} \sqcup Z_R,$$

and edges: rc whenever $r \in \mathcal{R}$ and $c \in \mathcal{C}$ conflict; rz_R whenever $r \in \mathcal{R}$ covers $z \in Z$; $z_L c$ whenever $c \in \mathcal{C}$ covers $z \in Z$. There are no edges between Z_L and Z_R.

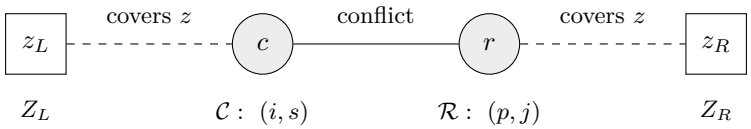


Figure 1: The three edges forced by one conflict between $r = (p, j)$ and $c = (i, s)$, where $z = (i, j)$. Different conflicts may share vertices and edges.

(A1) A conflict-free member is isolated in Γ. Suppose $(p, j) \in \mathcal{R}$ covers $(i, j) \in Z$, so $p \to i$. A conflict witnessing $(i, j) \in Z$ supplies $(i, s) \in \mathcal{C}$ with $s \to j$, and this member conflicts with (p, j). The case of $\mathcal{C}$ is symmetric.

(A2) A small cover suffices. Let K be a vertex cover of Γ with $|K| \le |Z| + |H|$ that contains no isolated vertex. Put $\mathcal{R}' = \mathcal{R} \setminus K$ and $\mathcal{C}' = \mathcal{C} \setminus K$. Every conflict is an edge with an endpoint in K, so no conflict survives. By (A1), conflict-free members are retained. Let q be the number of points of Z covered by no member of $\mathcal{R}' \cup \mathcal{C}'$. Each of the other $|Z| - q$ points z is covered by a surviving member, which is adjacent to a copy of z and is not in K; so that copy lies in K. Distinct points give distinct copies, whence

$$|K \cap (\mathcal{R} \sqcup \mathcal{C})| \le |K| - (|Z| - q) \le |H| + q, \tag{11}$$

which is (10). It remains to find K.

Part B: a special maximum matching. Among all maximum matchings of Γ choose one, $\mathcal{M}$, with the fewest edges between $\mathcal{R}$ and $\mathcal{C}$. Let k be the number of these edges, α the set of $\mathcal{M}$-edges between $\mathcal{R}$ and Z_R, and β those between Z_L and $\mathcal{C}$. With $\ell = |\alpha|$ and $t = |\beta|$,

$$|\mathcal{M}| = k + \ell + t.$$

Let $R_\alpha \subseteq \mathcal{R}$ and $C_\beta \subseteq \mathcal{C}$ be the sets of original endpoints of α and β. The endpoints of the k edges between $\mathcal{R}$ and $\mathcal{C}$ lie outside $R_\alpha \cup C_\beta$.

(B1) α is a maximum matching of the subgraph Γ_α induced on $\mathcal{R} \sqcup Z_R$, and β of the subgraph Γ_β induced on $Z_L \sqcup \mathcal{C}$. Suppose Γ_α has a larger matching. Its symmetric difference with α contains an α-augmenting path π in Γ_α, from an α-unmatched $r_0 \in \mathcal{R}$ to an α-unmatched $z_R \in Z_R$. Vertices of Z_R have neighbours only in $\mathcal{R}$, so their $\mathcal{M}$-edges lie in α; hence z_R is $\mathcal{M}$-unmatched. Each internal vertex of π is matched in $\mathcal{M}$ by an α-edge of π. If r_0 is $\mathcal{M}$-unmatched, π augments $\mathcal{M}$, contradicting maximality. Otherwise the $\mathcal{M}$-edge at r_0 is some $r_0 c$ with $c \in \mathcal{C}$. Then π is

augmenting for $\mathcal{M} \setminus \{r_0 c\}$, and augmenting gives a matching of size $|\mathcal{M}|$ with $k-1$ edges between $\mathcal{R}$ and $\mathcal{C}$, contradicting the choice of $\mathcal{M}$. The argument for β is the same with the sides exchanged.

Part C: four matrices with $NL = AB$. For each arrow $x \to y$ take two symbols a_{xy} and b_{xy}; set $a_{xy} = b_{xy} = 0$ when $x \not\to y$. For any real values of the symbols define

$$\begin{array}{ccc} \mathbb{R}^{\mathcal{R}} & \xrightarrow{L} & \mathbb{R}^{Z} \\ {\scriptstyle B}\downarrow & & \downarrow{\scriptstyle N} \\ \mathbb{R}^{H} & \xrightarrow{A} & \mathbb{R}^{\mathcal{C}} \end{array}$$

by (row index first, column index second)

$$\begin{aligned} L_{(i,j),(p,j)} &= a_{pi}, \quad N_{(i,s),(i,j)} = b_{sj}, \\ B_{(p,s),(p,j)} &= b_{sj}, \quad A_{(i,s),(p,s)} = a_{pi}, \end{aligned} \tag{12}$$

all entries not of a listed form being zero. So L and A move the first coordinate along $p \to i$ and fix the second, while N and B move the second coordinate from j to s along $s \to j$ and fix the first. In particular $L_{z,r} \neq 0$ only if r covers z, and $N_{c,z} \neq 0$ only if c covers z.

Fix a column $(p,j) \in \mathcal{R}$ and a row $(i,s) \in \mathcal{C}$. In $(NL)_{(i,s),(p,j)} = \sum_{z \in Z} N_{(i,s),z} L_{z,(p,j)}$ the only index z that can contribute is (i,j). In $(AB)_{(i,s),(p,j)}$ the only index of H that can contribute is (p,s). If (p,j) and (i,s) conflict, then $(i,j) \in Z$ and $(p,s) \in H$, and both entries equal $a_{pi} b_{sj}$. If they do not conflict, then $p \not\to i$ or $s \not\to j$, so $a_{pi} b_{sj} = 0$ and both entries vanish. Hence

$$NL = AB. \tag{13}$$

Part D: three conditions that give the matching bound. Suppose real values are chosen with

$$\operatorname{rank} L = \ell, \qquad \operatorname{rank} N = t, \qquad \dim \ker A \geq k. \tag{14}$$

Then

$$|H| = \dim \ker A + \operatorname{rank} A \geq k + \operatorname{rank}(AB) = k + \operatorname{rank}(NL) \geq k + \ell + t - |Z|. \tag{15}$$

The steps are: rank–nullity for A; $\operatorname{im}(AB) \subseteq \operatorname{im} A$; (13); and $\operatorname{rank}(NL) = \dim \operatorname{im} L - \dim(\operatorname{im} L \cap \ker N) \geq \ell - (|Z| - t)$. So (14) implies $|\mathcal{M}| \leq |Z| + |H|$. Part E obtains the first two conditions and Parts F–H the third.

Part E: one real assignment for all rank conditions. For every arrow $p \to i$ define

$$J_p = \{j : (p,j) \in \mathcal{R}\}, \qquad S_i = \{s : (i,s) \in \mathcal{C}\}, \qquad W^{p,i} = (b_{sj})_{s \in S_i,\ j \in J_p},$$

a matrix supported on $\Omega^{p,i} = \{(s,j) \in S_i \times J_p : s \to j\}$. Let $\rho^{p,i}$ be the maximum rank of a real matrix supported on $\Omega^{p,i}$.

Regard the symbols a_{xy}, b_{xy} $(x \to y)$ as indeterminates, so that L, N^{T}, $W^{p,i}$ have entries in the polynomial ring $\mathbb{R}[a,b]$. The idea is the classical link between generic rank and maximum matchings [9, Section 5, Theorem 1]. Because indeterminates are shared between matrices, we argue directly. Consider these finitely many polynomials.

(P1) π_α: the minor of L on the columns R_α and the rows z with z_R matched by α.

(P2) π_β: the minor of N^{T} (columns indexed by $\mathcal{C}$, rows by Z) on the columns C_β and the rows z with z_L matched by β.

(P3) For each arrow $p \to i$, a $\rho^{p,i} \times \rho^{p,i}$ minor $\pi^{p,i}$ of $W^{p,i}$, chosen below.

(P1) is a nonzero polynomial. The entry of L in row (i, j') and column (p, j) vanishes unless $j' = j$, and α matches each (p, j) to a point with the same second coordinate. Ordering rows and columns by second coordinate, the submatrix is block diagonal with one *square* block per j, so π_α is $\pm$ the product of the block determinants. In the block for j, the entry in row (i, j) and column (p, j) is the indeterminate a_{pi} if $p \to i$ and 0 otherwise, and distinct positions carry distinct indeterminates. So each permutation contributing to the block determinant gives $\pm$ a squarefree monomial that determines the permutation, and no two terms cancel. The restriction of α to the block is one such permutation. Hence each block determinant is nonzero. An indeterminate may occur in several blocks, but $\mathbb{R}[a, b]$ is an integral domain, so $\pi_\alpha \neq 0$.

(P2) is a nonzero polynomial. The same argument applies: $(N^{\mathsf{T}})_{(i,j),(i',s)}$ vanishes unless $i' = i$, the blocks are indexed by the first coordinate i, and within a block distinct positions carry distinct indeterminates b_{sj}.

(P3) can be chosen nonzero. Take a real matrix W_0 supported on $\Omega^{p,i}$ of rank $\rho^{p,i}$, and a nonzero $\rho^{p,i} \times \rho^{p,i}$ minor of W_0 (the empty minor if $\rho^{p,i} = 0$). Let $\pi^{p,i}$ be the same minor of $W^{p,i}$. The entries of $W^{p,i}$ on $\Omega^{p,i}$ are distinct indeterminates, so substituting the entries of W_0 is a legitimate evaluation of $\pi^{p,i}$. It gives a nonzero number, hence $\pi^{p,i} \neq 0$.

One assignment. The product of π_α, π_β and all $\pi^{p,i}$ is a nonzero element of $\mathbb{R}[a, b]$, again because $\mathbb{R}[a, b]$ is a domain. A nonzero real polynomial is nonzero at some real point (induction on the number of variables: a nonzero one-variable polynomial has finitely many roots). Fix such a point once and for all. From now on the symbols are these real numbers, and every polynomial in (P1)–(P3) is nonzero at them. Consequences:

(E1) $\operatorname{rank} L = \ell$, and the columns R_α of L form a basis of $\operatorname{im} L$. For the upper bound, a nonzero square minor of L has a nonzero term in its expansion. That term selects, for the chosen columns r, distinct rows z with $L_{z,r} \neq 0$, i.e. edges rz_R of Γ_α; this is a matching of the size of the minor, which is at most ℓ by (B1). For the lower bound, $\pi_\alpha \neq 0$ makes the ℓ columns R_α independent.

(E2) Likewise $\operatorname{rank} N = \operatorname{rank} N^{\mathsf{T}} = t$, and the columns C_β of N^{T} form a basis of $\operatorname{im} N^{\mathsf{T}}$.

(E3) Each $W^{p,i}$ has rank $\rho^{p,i}$: at least $\rho^{p,i}$ since $\pi^{p,i} \neq 0$, at most $\rho^{p,i}$ since it is supported on $\Omega^{p,i}$. So $W^{p,i}$ has maximum rank among real matrices supported on $\Omega^{p,i}$.

Part F: lifts. Let rc be an edge of $\mathcal{M}$ with $r \in \mathcal{R}$, $c \in \mathcal{C}$; then $r \notin R_\alpha$ and $c \notin C_\beta$. By (E1), Le_r is a combination of the columns R_α of L. By (E2), $N^{\mathsf{T}} e_c$ is a combination of the columns C_β of N^{T}. So there are reals $\lambda^r_\rho, \mu^c_\gamma$ with

$$u^r = e_r + \sum_{\rho \in R_\alpha} \lambda^r_\rho e_\rho \in \ker L, \qquad v^c = e_c + \sum_{\gamma \in C_\beta} \mu^c_\gamma e_\gamma \in \ker N^{\mathsf{T}}. \tag{16}$$

If r, r' are left endpoints of two such edges, the r'-coordinate of u^r is 1 if $r' = r$ and 0 otherwise. Also $v^c_c = 1$.

Part G: the kernel restriction. We claim that

$$u_{(p,j)}\, v_{(i,s)} = 0 \quad \text{for every conflict } ((p, j), (i, s)), \text{ every } u \in \ker B, \text{ and every } v \in \ker N^{\mathsf{T}}. \tag{17}$$

Fix such a conflict, u and v. Then $p \to i$, and we write $W = W^{p,i}$.

(G1) $x = (u_{(p,j')})_{j' \in J_p}$ *lies in* $\ker W$. The row of Wx at $s' \in S_i$ is $\sum_{j' \in J_p} b_{s'j'} u_{(p,j')}$. Suppose first that $s' \to j'$ for some $j' \in J_p$. Then $(p, j') \in \mathcal{R}$ and $(i, s') \in \mathcal{C}$ conflict (as $p \to i$), so $(p, s') \in H$. By (12) the nonzero entries of B in row (p, s') sit in the columns (p, j'), $j' \in J_p$, with values $b_{s'j'}$. So the sum is the full (p, s')-coordinate of Bu, which is 0. If instead there is no such j', every coefficient $b_{s'j'}$ is 0 by convention and the sum is trivially 0; here (p, s') need not belong to H, and no coordinate of Bu is used.

(G2) $y = (v_{(i,s')})_{s' \in S_i}$ *lies in* $\ker W^{\mathsf{T}}$. The entry of $W^{\mathsf{T}} y$ at $j' \in J_p$ is $\sum_{s' \in S_i} b_{s'j'} v_{(i,s')}$. If $s' \to j'$ for some $s' \in S_i$, the same conflict gives $(i, j') \in Z$, and the sum is the full (i, j')-coordinate of $N^{\mathsf{T}} v = 0$. Otherwise all coefficients vanish.

(G3) We have $(s, j) \in \Omega^{p,i}$, because $s \in S_i$, $j \in J_p$ and $s \to j$. By (E3), Lemma 3.2 applies to the fixed real matrix W and gives $x_j y_s = 0$, which is (17). The perturbation inside the proof of that lemma is only a comparison matrix supported on $\Omega^{p,i}$; the fixed assignment is not changed.

Part H: $\dim \ker A \geq k$. Let rc be one of the k edges of $\mathcal{M}$ between $\mathcal{R}$ and $\mathcal{C}$. It is a conflict, so (17) with $v = v^c$ and $v^c_c = 1$ shows that *every* $u \in \ker B$ has $u_r = 0$. Now suppose $\sum_r \theta_r B u^r = 0$, the sum over the k left endpoints. Then $u = \sum_r \theta_r u^r \in \ker B$, and its coordinate at each such endpoint r is θ_r by Part F; hence all $\theta_r = 0$. So the k vectors Bu^r are linearly independent. They lie in $\ker A$, since $A(Bu^r) = NLu^r = 0$ by (13) and (16). With (E1) and (E2), all three conditions in (14) hold, and (15) gives

$$|\mathcal{M}| = k + \ell + t \leq |Z| + |H|.$$

Part I: from the matching to the cover. This is the standard argument for Kőnig's theorem [17]. Starting from all $\mathcal{M}$-unmatched vertices of $\mathcal{V}_L$, follow edges outside $\mathcal{M}$ from left to right and edges of $\mathcal{M}$ from right to left. Let W_L, W_R be the sets of reached vertices on the two sides. No unmatched right vertex is reached, since the path to it would augment $\mathcal{M}$. The endpoints of a matched edge are both reached or both unreached: a reached right endpoint leads to its mate, and a matched left vertex can be reached only through its mate. Put

$$K = (\mathcal{V}_L \setminus W_L) \cup W_R.$$

K is a cover: an edge from W_L to $\mathcal{V}_R \setminus W_R$ can be neither unmatched (it would be traversed) nor matched (its endpoints have equal reachability). K contains exactly one endpoint of each matched edge and no unmatched vertex, so $|K| = |\mathcal{M}| \leq |Z| + |H|$. Isolated vertices are unmatched and hence not in K. By (A2) the lemma follows. □

4 The extremal contradiction

Proposition 4.1. *There is no nonempty finite oriented graph G in which every vertex has an in-neighbor and whose one-step image operator*

$$F(U) = \{y \in V(G) : x \to y \text{ for some } x \in U\}$$

satisfies

$$|U| + |F^2(U)| < 2|F(U)| \qquad (\varnothing \neq U \subsetneq V(G)), \tag{18}$$

where $F^2 = F \circ F$.

Proof. Plan. Assume G exists. Choose an extremal compatible pair (P, Q) (Steps 1–2). Use (18) to build larger families $\mathcal{R}, \mathcal{C}$ (Steps 3–4). Check the hypotheses needed to prune them cheaply (Steps 5–6). Prune, and beat the extremal value (Step 7).

Step 1: coordinate images and compatibility. Put $X = V(G)$. For $\mathcal{S} \subseteq X \times X$ define

$$F_1\mathcal{S} = \{(i,j) : (p,j) \in \mathcal{S} \text{ and } p \to i \text{ for some } p \in X\},$$
$$F_2\mathcal{S} = \{(i,j) : (i,s) \in \mathcal{S} \text{ and } s \to j \text{ for some } s \in X\},$$

$F_1^2 = F_1 \circ F_1$, $F_2^2 = F_2 \circ F_2$, and $\Delta = \{(v,v) : v \in X\}$. Call $P, Q \subseteq X \times X$ *compatible* if $F_1P \cap F_2Q = \varnothing$. Unwinding the definitions, a point (i,j) lies in $F_1P \cap F_2Q$ exactly when some $(p,j) \in P$ and $(i,s) \in Q$ satisfy $p \to i$ and $s \to j$. So compatibility means that P (as $\mathcal{R}$) and Q (as $\mathcal{C}$) have no conflict in the sense of Lemma 3.1. Likewise, F_1P is the set of points covered by a member of P, and F_2Q the set covered by a member of Q.

Step 2: the extremal pair. The pair (Δ, Δ) is compatible: $(i,j) \in F_1\Delta$ means $j \to i$ and $(i,j) \in F_2\Delta$ means $i \to j$, and both cannot hold in an oriented graph. As X is finite, we may choose a compatible pair (P, Q) with $\Delta \subseteq P \cap Q$ maximizing $M + d$, where

$$M = |P| + |Q|, \qquad d = |X|^2 - |F_1P| - |F_2Q|.$$

By compatibility, d is the number of points outside $F_1P \cup F_2Q$.

Step 3: applying (18) *fibrewise.* For $j \in X$ let $P_j = \{p : (p,j) \in P\}$ be the j-th column of P. By definition of F_1, the j-th column of F_1P is $F(P_j)$, and that of F_1^2P is $F^2(P_j)$.

P_j is nonempty since $j \in P_j$. It is proper. Indeed, positive indegrees give $F(X) = X$, so $P_j = X$ would give $X \times \{j\} \subseteq F_1P$. Choose an in-neighbour w of j; then $(w,w) \in Q$ and $w \to j$ give $(w,j) \in F_2Q$, contradicting compatibility. Symmetrically, each row $Q_i = \{s : (i,s) \in Q\}$ contains i. If $Q_i = X$ then $\{i\} \times X \subseteq F_2Q$, while $w \to i$ and $(w,w) \in P$ give $(i,w) \in F_1P$, again a contradiction.

Summing (18) for $U = P_j$ over $j \in X$, and for $U = Q_i$ over $i \in X$, gives

$$|P| + |F_1^2P| < 2|F_1P|, \qquad |Q| + |F_2^2Q| < 2|F_2Q|. \tag{19}$$

Step 4: the complement families. Define

$$\mathcal{R} = (X \times X) \setminus F_2^2Q, \qquad \mathcal{C} = (X \times X) \setminus F_1^2P.$$

By (19),

$$|\mathcal{R}| + |\mathcal{C}| = 2|X|^2 - |F_2^2Q| - |F_1^2P| > 2|X|^2 + M - 2(|F_1P| + |F_2Q|) = M + 2d. \tag{20}$$

Step 5: the diagonal lies in both families and is conflict-free. Membership. If $(v,v) \in F_2^2Q$, there is $s \to v$ with $(v,s) \in F_2Q$; but $(s,s) \in P$ and $s \to v$ give $(v,s) \in F_1P$, contradicting compatibility. If $(v,v) \in F_1^2P$, there is $p \to v$ with $(p,v) \in F_1P$; but $(p,p) \in Q$ and $p \to v$ give $(p,v) \in F_2Q$.

No conflicts. A conflict between $(v,v) \in \mathcal{R}$ and $(i,s) \in \mathcal{C}$ means $v \to i$ and $s \to v$. Then $(s,s) \in P$ gives $(v,s) \in F_1P$ and $(i,s) \in F_1^2P$, contradicting $(i,s) \in \mathcal{C}$. A conflict between $(p,j) \in \mathcal{R}$ and $(v,v) \in \mathcal{C}$ means $p \to v$ and $v \to j$. Then $(p,p) \in Q$ gives $(p,v) \in F_2Q$ and $(p,j) \in F_2^2Q$, contradicting $(p,j) \in \mathcal{R}$.

Step 6: $|H| \le d$. Let H, Z be the sets attached to $\mathcal{R}, \mathcal{C}$ in Lemma 3.1. We claim

$$H \subseteq (X \times X) \setminus (F_1P \cup F_2Q), \qquad \text{hence} \qquad |H| \le d. \tag{21}$$

Take a conflict $(p,j) \in \mathcal{R}$, $(i,s) \in \mathcal{C}$ with $p \to i$ and $s \to j$. If $(p,s) \in F_1 P$, then $p \to i$ puts (i,s) in $F_1^2 P$, contradicting $(i,s) \in \mathcal{C}$. If $(p,s) \in F_2 Q$, then $s \to j$ puts (p,j) in $F_2^2 Q$, contradicting $(p,j) \in \mathcal{R}$.

Step 7: pruning and the contradiction. Lemma 3.1 gives conflict-free $P' \subseteq \mathcal{R}$ and $Q' \subseteq \mathcal{C}$ retaining all conflict-free members. By Step 5, $\Delta \subseteq P' \cap Q'$. By Step 1, (P', Q') is compatible. So (P', Q') competes in the maximization of Step 2, with

$$d' = |X|^2 - |F_1 P'| - |F_2 Q'|$$

equal to the number of points of $X \times X$ outside $F_1 P' \cup F_2 Q'$. By Step 1, a point is covered by a member of $P' \cup Q'$ exactly when it lies in $F_1 P' \cup F_2 Q'$. So the uncovered points of Z in (10) are among these d' points. With (21), at most $d + d'$ members were deleted, and (20) gives

$$|P'| + |Q'| + d' \geq |\mathcal{R}| + |\mathcal{C}| - (d + d') + d' > M + 2d - (d + d') + d' = M + d.$$

This contradicts the choice of (P, Q). □

Proof of Theorem 1.1. If the theorem were false, Proposition 2.1 would give a nonempty finite oriented graph with positive indegrees satisfying (18) for every nonempty proper vertex set. Proposition 4.1 excludes this. □

5 Weighted consequences

The corollary applies Theorem 1.1 through Seacrest's established equivalences [19], which are not new here. Part (iii) is taken from that reference without re-derivation. Parts (i) and (ii) are then checked directly.

Corollary 5.1 (Weighted second-neighborhood forms)**.** *Let D be a nonempty finite oriented graph with arc set $A(D)$. For $\eta : V(D) \to [0, \infty)$ write $\eta(S) = \sum_{x \in S} \eta(x)$.*

(i) *For every vertex weighting $\eta : V(D) \to [0, \infty)$ there is a vertex v with $\eta(N_1^+(v)) \leq \eta(N_2^+(v))$.*

(ii) *There is a vertex weighting $\eta : V(D) \to [0, \infty)$ with $\sum_x \eta(x) = 1$ such that $\eta(N_1^+(v)) \leq \eta(N_2^+(v))$ for every $v \in V(D)$. Some entries of η may be zero.*

(iii) *For every arc weighting $w : A(D) \to [0, \infty)$, set $w(vs) = 0$ when $vs \notin A(D)$, and for $v, s \in V(D)$ put*

$$\beta_v^w(s) = \max\Big(\{0\} \cup \{w(us) - w(vs) : u \in V(D),\ v \to u \to s\}\Big).$$

Then there is a vertex v with

$$\sum_{s \in V(D)} \beta_v^w(s) \geq \sum_{u \in N_1^+(v)} w(vu).$$

Here $w(vs)$ enters only as a number. Whether $v \to u \to s$ is a path is decided by the arcs of D: an arc of weight zero is still an arc, and $\beta_v^w(s) = 0$ if there is no such path.

Proof. (iii) According to the source, Seacrest's Proposition 2 [19] proves that his arc-weighted Conjecture 4, which is statement (iii) for all nonempty finite oriented graphs, is equivalent to the unweighted conjecture. Theorem 1.1 therefore gives (iii).

(i) Apply (iii) with $w(us) = \eta(s)$ on every arc $u \to s$. Fix v, s. If $v \to s$, every eligible difference is $\eta(s) - \eta(s) = 0$, so $\beta_v^w(s) = 0$. If $s \in N_2^+(v)$, then $w(vs) = 0$ and a path $v \to u \to s$ exists, so $\beta_v^w(s) = \eta(s)$. In all other cases (including $s = v$, by orientation) there is no path $v \to u \to s$ and $\beta_v^w(s) = 0$. So the two sides of (iii) are $\eta(N_2^+(v))$ and $\eta(N_1^+(v))$. This is the specialization noted after Seacrest's Proposition 2.

(ii) Let Φ_D be the $V(D) \times V(D)$ matrix with entry 1 at (v, s) when $s \in N_2^+(v)$, -1 when $s \in N_1^+(v)$, and 0 otherwise. Then $(\Phi_D \eta)_v = \eta(N_2^+(v)) - \eta(N_1^+(v))$. Let $\overleftarrow{D}$ be D with all arcs reversed. Since s is an exact second out-neighbour of v in $\overleftarrow{D}$ exactly when $v \in N_2^+(s)$ in D, and similarly for first neighbours, $\Phi_{\overleftarrow{D}} = \Phi_D^{\mathsf{T}}$. The source cites Seacrest's Theorem 1, which applies Farkas' lemma, for the following alternative: for a real matrix Φ, either (a) some nonzero $\eta \geq 0$ has $\Phi\eta \geq 0$, or (b) some $\xi \geq 0$ has $\Phi^{\mathsf{T}}\xi < 0$ in every coordinate. One way to see it: by the minimax theorem, $\max_\eta \min_v (\Phi\eta)_v = \min_\xi \max_s (\Phi^{\mathsf{T}}\xi)_s$ over probability vectors; (a) holds if this value is ≥ 0 and (b) if it is < 0. For $\Phi = \Phi_D$, alternative (b) says $\xi(N_2^+(v)) < \xi(N_1^+(v))$ for every vertex v of $\overleftarrow{D}$, contradicting (i) for the oriented graph $\overleftarrow{D}$. So (a) holds; divide η by its total weight. □

6 Directed-cycle consequences

In an oriented graph with no directed triangle, every exact second out-neighbour is a non-neighbour. Part (i) below is the oriented-graph form of Thomassé's non-neighbor statement recorded by Sullivan [21, Conjecture 6.12]. Part (ii) is noted in [21, Section 3]. Part (iii) is the girth-four lower bound in the Behzad–Chartrand–Wall minimum-order conjecture [1], also recorded in [21, Section 4].

Corollary 6.1 (Non-neighbors and directed cycles)**.** *Let D be a nonempty finite oriented graph of order n. For $v \in V(D)$, write*

$$N_1^-(v) = \{u : u \to v\}, \qquad d^{\pm}(v) = |N_1^{\pm}(v)|, \qquad \delta^{\pm}(D) = \min_{x \in V(D)} d^{\pm}(x),$$

and let the set of non-neighbors of v, excluding v itself, be

$$\overline{N}(v) = V(D) \setminus \big(\{v\} \cup N_1^+(v) \cup N_1^-(v)\big).$$

(i) *If D has no directed 3-cycle, then some vertex v satisfies*

$$d^+(v) \leq |\overline{N}(v)|, \qquad n - 1 \geq 2d^+(v) + d^-(v).$$

(ii) *If $\delta^+(D) \geq n/3$ and $\delta^-(D) \geq n/3$, then D contains a directed 3-cycle.*

(iii) *If $r \geq 1$ is an integer, $d^+(v) = d^-(v) = r$ for every vertex, and D has directed girth exactly four, then $n \geq 3r + 1$.*

Here directed girth is the length of a shortest directed cycle.

Proof. (i) Suppose D has no directed 3-cycle, and let $w \in N_2^+(v)$ with a path $v \to u \to w$. By definition $w \neq v$ and $w \notin N_1^+(v)$. Also $w \notin N_1^-(v)$, since $w \to v$ would close a directed 3-cycle.

Thus $N_2^+(v) \subseteq \overline{N}(v)$ for every v, and Theorem 1.1 gives a vertex with $d^+(v) \le |N_2^+(v)| \le |\overline{N}(v)|$. As D is oriented, $N_1^+(v)$ and $N_1^-(v)$ are disjoint, so

$$n - 1 = d^+(v) + d^-(v) + |\overline{N}(v)| \ge 2d^+(v) + d^-(v).$$

(ii) Without a directed 3-cycle, (i) would give $n - 1 \ge 2\delta^+(D) + \delta^-(D) \ge n$, which is absurd.
(iii) Directed girth four excludes directed 3-cycles, and (i) with all degrees equal to r reads $n - 1 \ge 3r$. □

7 Verification checklist

No step of the rewrite is marked unverified. A sceptical reader should check these load-bearing points first.

1. *Lemma 2.2, Steps 6 and 8.* The inclusion $N^+_{2,D_T}(v) \subseteq (N^+_{2,D}(v) \setminus C) \cup (Q \cap F(T))$ and the *exact* identity $g(T \cup Q) - g(T) = |C| - |Q \cap F(T)|$. Together with $d_2 < d_1$ they give the strict increment (7).

2. *Lemma 2.3, Steps 5–7.* The upper bound for $|F^2_G(U)|$, the identity $|H_0| - 2|F(S)| + |S \cap F(S)| = |F^2(S) \setminus F(S)| - |F(S) \setminus S|$, and the separate treatment of $S = V(D)$.

3. *Lemma 3.1, Part A.* Inequality (11): every covered point of Z forces a copy into the cover. This is where the "uncovered" term of (10) arises.

4. *Lemma 3.1, Parts C–D.* The identity $NL = AB$ (13), checked entrywise, and the chain (15).

5. *Lemma 3.1, Part E.* Each of (P1)–(P3) is a nonzero polynomial despite shared indeterminates, and one real point makes them all nonzero. The bound $\operatorname{rank} L \le \ell$ uses (B1).

6. *Lemma 3.1, Parts G–H.* In (G1)–(G2) each sum is either a full coordinate of Bu or $N^{\mathsf{T}}v$, or has all coefficients zero. Lemma 3.2 then gives (17), which forces the Bu^r to be independent.

7. *Proposition 4.1, Steps 5–7.* The diagonal members are conflict-free (so they survive), $|H| \le d$ (21), the uncovered points of Z number at most d', and the final arithmetic $M + 2d - (d + d') + d' = M + d$.

8. *External inputs.* Theorem 1.1 uses no cited result beyond standard matching theory. Corollary 5.1(iii) rests entirely on Seacrest's Proposition 2 [19], quoted as reported by the source and not re-derived here; compare the definition of β_v^w with that reference.

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